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WEEK2 ASSIGNMENT

2.3

1. Creating a database and a table
 - a. Create a database "academic"

```
MariaDB [(none)]> create database academic;  
Query OK, 1 row affected (0.001 sec)  
  
MariaDB [(none)]> show databases;  
+-----+  
| Database |  
+-----+  
| academic |  
| information_schema |  
| mysql |  
| performance_schema |  
| phpmyadmin |  
| test |  
+-----+  
6 rows in set (0.001 sec)
```

- b. Create a table "student"

```

MariaDB [(none)]> use academic;
Database changed
MariaDB [academic]> create table student (
  -> NIM INT AUTO_INCREMENT NOT NULL,
  -> name VARCHAR(255),
  -> gender VARCHAR(20),
  -> birthplace VARCHAR(255),
  -> telephone VARCHAR(20),
  -> academic_advisor VARCHAR(255),
  -> PRIMARY KEY (NIM)
  -> );
Query OK, 0 rows affected (0.011 sec)

MariaDB [academic]> describe student;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| NIM   | int(11) | NO | PRI | NULL | auto_increment |
| name  | varchar(255) | YES | | NULL | |
| gender | varchar(20) | YES | | NULL | |
| birthplace | varchar(255) | YES | | NULL | |
| telephone | varchar(20) | YES | | NULL | |
| academic_advisor | varchar(255) | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.013 sec)

```

c. Create a table "lecturer"

```

MariaDB [academic]> create table lecturer (
  -> NIP INT AUTO_INCREMENT NOT NULL,
  -> name VARCHAR(255),
  -> gender VARCHAR(20),
  -> position VARCHAR(255),
  -> interest TEXT,
  -> telephone VARCHAR(20),
  -> PRIMARY KEY (NIP)
  -> );
Query OK, 0 rows affected (0.009 sec)

MariaDB [academic]> describe lecturer;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| NIP   | int(11) | NO | PRI | NULL | auto_increment |
| name  | varchar(255) | YES | | NULL | |
| gender | varchar(20) | YES | | NULL | |
| position | varchar(255) | YES | | NULL | |
| interest | text | YES | | NULL | |
| telephone | varchar(20) | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.014 sec)

```

d. Create a table "course"

```

MariaDB [academic]> create table course (
  -> course_id INT AUTO_INCREMENT NOT NULL,
  -> course_name VARCHAR(255),
  -> SKS INT,
  -> subject_lecturer VARCHAR(255),
  -> course_date DATETIME,
  -> course_class VARCHAR(50),
  -> PRIMARY KEY (course_id)
  -> );
Query OK, 0 rows affected (0.011 sec)

MariaDB [academic]> describe course;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| course_id | int(11) | NO | PRI | NULL | auto_increment |
| course_name | varchar(255) | YES | | NULL | |
| SKS | int(11) | YES | | NULL | |
| subject_lecturer | varchar(255) | YES | | NULL | |
| course_date | datetime | YES | | NULL | |
| course_class | varchar(50) | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.016 sec)

```

e. Create a table “KRS”

```

MariaDB [academic]> create table KRS (
  -> KRS_id INT AUTO_INCREMENT NOT NULL,
  -> course_id VARCHAR(20),
  -> NIM INT,
  -> year YEAR,
  -> semester INT,
  -> grade DECIMAL(5,2),
  -> PRIMARY KEY (KRS_id)
  -> );
Query OK, 0 rows affected (0.010 sec)

MariaDB [academic]> describe KRS;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| KRS_id | int(11) | NO | PRI | NULL | auto_increment |
| course_id | varchar(20) | YES | | NULL | |
| NIM | int(11) | YES | | NULL | |
| year | year(4) | YES | | NULL | |
| semester | int(11) | YES | | NULL | |
| grade | decimal(5,2) | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.015 sec)

```

2. I have shown all the table in the previous task number
3. I have shown all the structure in the table in the previous task number
4. Creating a query
 - a. Drop primary key in “NIP”

```

MariaDB [academic]> ALTER TABLE lecturer DROP PRIMARY KEY;
ERROR 1075 (42000): Incorrect table definition; there can be only one auto column and it must be defined as a key
MariaDB [academic]> ALTER TABLE lecturer
-> CHANGE COLUMN NIP NIP INT NOT NULL;
Query OK, 0 rows affected (0.007 sec)
Records: 0 Duplicates: 0 Warnings: 0

MariaDB [academic]> ALTER TABLE lecturer DROP PRIMARY KEY;
Query OK, 0 rows affected (0.030 sec)
Records: 0 Duplicates: 0 Warnings: 0

```

To drop a primary key, First I have to alter the table and change the “NIP” data type and remove the auto increment or it will give an error. After that, I can drop the primary key.

- b. Add primary key to “NIP”

```

MariaDB [academic]> ALTER TABLE lecturer
-> ADD PRIMARY KEY (NIP);
Query OK, 0 rows affected (0.023 sec)
Records: 0 Duplicates: 0 Warnings: 0

```

To add the primary key, I just need to alter the table and add a primary key to the attribute that I want.

- c. Look for lecture table

```

MariaDB [academic]> describe lecturer;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| NIP   | int(11) | NO | PRI | NULL | auto_increment |
| name  | varchar(255) | YES | | NULL | |
| gender | varchar(20) | YES | | NULL | |
| position | varchar(255) | YES | | NULL | |
| interest | text | YES | | NULL | |
| telephone | varchar(20) | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.014 sec)

```

- d. Change table name “lecturer” to “professor”

```

MariaDB [academic]> RENAME TABLE lecturer to professor;
Query OK, 0 rows affected (0.009 sec)

```

- e. In “professor” table, change attribute “name” to “lecturer_name”

```
MariaDB [academic]> ALTER TABLE professor
    -> change name lecturer_name VARCHAR(255);
Query OK, 0 rows affected (0.007 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

- f. Change data type of attribute “gender” to enum('man','woman')

```
MariaDB [academic]> ALTER TABLE student
    -> change gender gender ENUM('man','woman');
Query OK, 0 rows affected (0.024 sec)
Records: 0 Duplicates: 0 Warnings: 0

MariaDB [academic]> ALTER TABLE lecturer
    -> change gender gender ENUM('man','woman');
ERROR 1146 (42S02): Table 'academic.lecturer' doesn't exist
MariaDB [academic]> ALTER TABLE professor
    -> change gender gender ENUM('man','woman');
Query OK, 0 rows affected (0.028 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

- g. Change data type of attribute ‘telephone’ to INT

```
MariaDB [academic]> ALTER TABLE student
    -> change telephone telephone INT;
Query OK, 0 rows affected (0.032 sec)
Records: 0 Duplicates: 0 Warnings: 0

MariaDB [academic]> ALTER TABLE professor
    -> change telephone telephone INT;
Query OK, 0 rows affected (0.053 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

2.4

1. Apa perbedaan antara Create, Alter dan Drop pada Data Definition Language?

Create → to create a new database or table.

Alter → to modify a structure in an existing table. You can use it to add, rename, or remove a column in a table.

Drop → to delete an existing database, table, or remove a column.

2. Perintah apa yang seharusnya digunakan untuk mendefinisikan sebuah objek dari Database?

You can use a command `USE [database]`

3. Alih-alih menghapus suatu database, apa perintah yang sebaiknya digunakan untuk merubah nama suatu database?

You can use a dumping and reimporting method

1. First make physical backups of database using mysqldump
`mysqldump -u username -p"password" -R oldDbName > oldDbName.sql`
2. Then create a new database
3. Lastly, import the dump file into new database
`mysql -u username -p"password" newDbName < oldDbName.sql`

2.5

1. Create a Hotel Database

```
MariaDB [(none)]> create database hotel;  
Query OK, 1 row affected (0.001 sec)
```

```
MariaDB [(none)]> show databases;
```

Database
academic
hotel
information_schema
mysql
performance_schema
phpmyadmin
test

```
7 rows in set (0.001 sec)
```

```
MariaDB [(none)]> use hotel;  
Database changed
```

2. Define room, guest, and reservation table with primary key and foreign key

```
MariaDB [hotel]> create table guest (  
    -> guest_id INT AUTO_INCREMENT NOT NULL,  
    -> name VARCHAR(255),  
    -> email VARCHAR(255),  
    -> address TEXT,  
    -> PRIMARY KEY (guest_id)  
    -> );  
Query OK, 0 rows affected (0.010 sec)
```

```
MariaDB [hotel]> create table room (  
    -> room_id INT AUTO_INCREMENT NOT NULL,  
    -> guest_id INT,  
    -> PRIMARY KEY (room_id),  
    -> FOREIGN KEY (guest_id) REFERENCES guest(guest_id)  
    -> );  
Query OK, 0 rows affected (0.017 sec)
```

```
MariaDB [hotel]> create table reservation (  
    -> reservation_id INT AUTO_INCREMENT NOT NULL,  
    -> room_id INT,  
    -> guest_id INT,  
    -> PRIMARY KEY (reservation_id),  
    -> FOREIGN KEY (room_id) REFERENCES room(room_id),  
    -> FOREIGN KEY (guest_id) REFERENCES guest(guest_id)  
    -> );  
Query OK, 0 rows affected (0.020 sec)
```

3. Add telephone column in guest table

```
MariaDB [hotel]> ALTER TABLE guest  
    -> ADD COLUMN telephone VARCHAR(20)  
    -> AFTER email;  
Query OK, 0 rows affected (0.005 sec)  
Records: 0 Duplicates: 0 Warnings: 0
```

4. Delete the reservation table

```
MariaDB [hotel]> DROP TABLE reservation;
Query OK, 0 rows affected (0.037 sec)

MariaDB [hotel]> describe reservation;
ERROR 1146 (42S02): Table 'hotel.reservation' doesn't exist
MariaDB [hotel]>
```

The result of Hotel database

```
MariaDB [hotel]> describe guest;
+-----+-----+-----+-----+-----+-----+
| Field      | Type      | Null | Key | Default | Extra      |
+-----+-----+-----+-----+-----+-----+
| guest_id   | int(11)   | NO   | PRI | NULL    | auto_increment |
| name       | varchar(255) | YES  |     | NULL    |              |
| email      | varchar(255) | YES  |     | NULL    |              |
| telephone  | varchar(20) | YES  |     | NULL    |              |
| address    | text      | YES  |     | NULL    |              |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.011 sec)

MariaDB [hotel]> describe room;
+-----+-----+-----+-----+-----+-----+
| Field      | Type      | Null | Key | Default | Extra      |
+-----+-----+-----+-----+-----+-----+
| room_id    | int(11)   | NO   | PRI | NULL    | auto_increment |
| guest_id   | int(11)   | YES  | MUL | NULL    |              |
+-----+-----+-----+-----+-----+-----+
2 rows in set (0.011 sec)

MariaDB [hotel]> describe reservation;
+-----+-----+-----+-----+-----+-----+
| Field      | Type      | Null | Key | Default | Extra      |
+-----+-----+-----+-----+-----+-----+
| reservation_id | int(11) | NO   | PRI | NULL    | auto_increment |
| room_id       | int(11) | YES  | MUL | NULL    |              |
| guest_id      | int(11) | YES  | MUL | NULL    |              |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.020 sec)
```